Business Requirements Document – TaskFlow Assistant (TFA)

TaskFlow – Intelligent Task Allocation & Tracking System

Document: Business Requirements Document (sub-module 08.1 Assistant, code `TFA`) · **Version:** 1.0 · **Status:** Approved
Project type: Personal Project (Skill Development) · **Date:** 31-Oct-2026
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Document Control

Revision History

Version	Date	Author	Summary of changes
1.0	31-Oct-2026	Jojo George	Initial version of the TFA sub-module BRD (chat, actions, how-to help, safeguards, voice input)
1.0	31-Oct-2026	Jojo George	Approved and signed off; baselined as the agreed business requirements.
1.0	31-Oct-2026	Jojo George	Editorial only: Related Documents paths updated to the `TF - Documents` folder layout; no requirement changed, sign-off unchanged.

Approvals / Sign-off

Role	Name	Decision	Date
Product owner	Jojo George	Approved	31-Oct-2026
Reviewer	Jojo George	Approved	31-Oct-2026

Related Documents

Document	Location
Overall BRD (roles, global rules, NFRs, Legal)	`../../00-Overall/BRD-TaskFlow-Overall.md`
BRD – TFA Settings (AI provider setup)	`../../06-Settings/06.3-TFA-Settings/BRD-TFA-Settings.md`
BRD – My Profile (profile rules used by TFA)	`../../07-My-Profile/07.1-My-Profile/BRD-My-Profile.md`
BRD – Mail (mail rules used by TFA)	`../../04-Communication/04.1-Mail/BRD-Mail.md`

Document	Location
FRD / HLD / LLD – Assistant	`../../../FRD/08-TFA-Assistant/08.1-Assistant/FRD-Assistant.md` and the matching HLD/LLD folders
TDD	`TF - Documents/TF - TDD/TDD-TaskFlow.md` (section 16)

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1. Project Overview

1.1 Introduction

TFA (TaskFlow Assistant) is the AI assistant built into TaskFlow. It opens as a panel on the right of every signed-in screen. Users ask how TaskFlow works, see their upcoming, due-today and overdue tasks, get a ranked "what to do first" list, and let TFA prepare a mail, a change to their own profile or (for administrators) a change to another user's role. TFA never acts on its own: every prepared action is shown on a card and happens only when the user clicks Confirm.

1.2 Business Problem / Current Situation

Users who do not know a screen must search training material or ask a colleague. Deciding what to work on first means scanning the task list by hand, and routine mails such as "update my manager on task TSK-12" take time to address and write. A general-purpose AI chat outside TaskFlow would not know the user's tasks, could leak company data, and could invent features or act without permission.

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1.3 Sub-module Objectives

ID	Objective	Success measure
OBJ-TFA-01	Give every permitted user one place on every screen to ask about TaskFlow and their own work	TFA opens from any signed-in screen and answers in the same conversation
OBJ-TFA-02	Give a clear, repeatable work order	The same open tasks always produce the same ranked list, whichever AI model is used
OBJ-TFA-03	Save time on routine actions without losing control	A mail or own-profile change is prepared and carried out with one confirmation
OBJ-TFA-04	Keep the organisation safe	TFA never does more than the user may do, never accepts passwords or keys, and records every action outcome
OBJ-TFA-05	Give trustworthy how-to answers	How-to answers come only from the 19 built-in help topics, with a link to the screen
OBJ-TFA-06	Allow hands-free use	In Chrome or Edge, "Hey TF assistant" opens TFA and answers are read aloud

1.4 Expected Benefits

- Less training effort; consistent how-to answers for every role.
- Faster daily planning through the ranked work order.
- Faster routine communication with mail drafts addressed to "my manager".
- The organisation chooses where AI runs: a local model (nothing leaves the machine) or a cloud model.

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2. Stakeholders & User Roles

2.1 Stakeholders

Stakeholder	Interest / responsibility
Project owner (Jojo George)	Defines priorities, approves this BRD
All TaskFlow users	Ask questions, plan work, confirm prepared actions
Administrators	Configure AI providers in TFA Settings; grant TFA permissions in Role Configuration; change roles through TFA

2.2 User Roles

Role	Description	Main responsibilities	Access level
Administrator	Full system user	Chat, mail drafts, own profile change, change another user's role	Own conversation and actions; role change across the organisation
Manager	Leads teams and projects	Chat, mail drafts, own profile change	Own conversation and actions; task answers follow own task access

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Role	Description	Main responsibilities	Access level
Employee	Works on assigned tasks	Chat, mail drafts, own profile change	Own conversation and actions; task answers follow own task access
Custom role	Role created in Role Configuration	As granted	As granted (`assistant.chat` , `assistant.actions.confirm` , `roles.manage`)

Permission	Administrator	Manager	Employee	Custom role
`assistant.chat` – chat with the assistant	Granted	Granted	Granted	Configurable
`assistant.actions.confirm` – confirm, edit or cancel prepared actions	Granted	Granted	Granted	Configurable
`roles.manage` (Role Configuration key, checked for role changes)	Granted	Denied	Denied	Configurable

3. Scope

3.1 In Scope

- TFA Panel: conversation, model selector, cloud notice, new chat, provider failure handling.
- Task answers: upcoming, due today, overdue, all open tasks; ranked "what to do first" list.
- Action Cards: mail draft (editable), own profile update, role change; confirm and cancel.
- Lookup and guidance: find a user, password guidance, "needs an administrator" guidance.
- How-to help from 19 built-in help topics, role-aware.
- Safeguards: password/API-key screen, 20-messages-per-minute limit, action audit log.
- Voice input: wake phrase, dictation, answers read aloud (Chrome, Edge).

3.2 Out of Scope (this release)

- AI provider, model and API key setup – TFA Settings (`../../06-Settings/06.3-TFA-Settings/BRD-TFA-Settings.md`).
- Creating, editing or reassigning tasks through TFA – not offered; TFA links to the task screens.
- Password changes through TFA – never; TFA points to My Profile.
- A screen to view the TFA audit log – not provided.
- Deleting one's own conversation history – not provided (see R-TFA-04).

3.3 Modules Overview

#	Module	Purpose	Screens
08.1	TFA Assistant	AI help, planning and confirm-before-act actions	SCREEN 1 TFA Panel (chat), SCREEN 2 How-to help (in panel), SCREEN 3 Action Card, SCREEN 4 Voice controls, SCREEN 5 Background safeguards (no screen)

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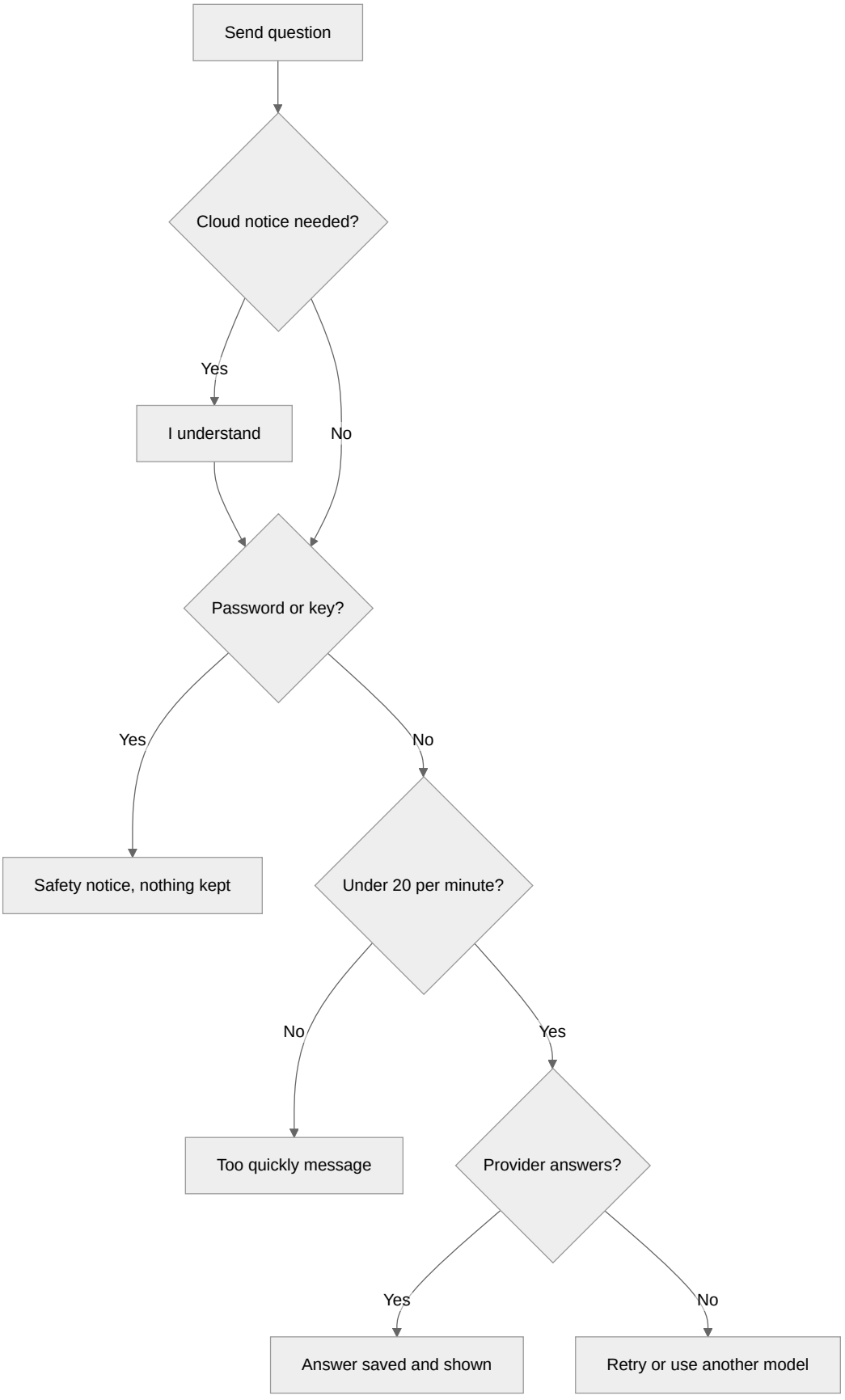
4. Business Process / Workflows

4.1 Ask TFA a Question

Trigger: User clicks the TFA button or says "Hey TF assistant" · **Actors:** User, System, AI provider

Step	Actor / System	Action	Result
1	User	Opens the panel and sends a question	Message held for checks
2	System	Shows the cloud notice for the first message to a cloud provider	User clicks **I understand** or **Cancel**
3	System	Checks length (1–2000) and screens for passwords or API keys	Unsafe message is neither sent nor stored
4	System	Checks the 20-messages-per-minute limit	Over the limit → "too quickly" message
5	AI provider	Answers, using up to 4 rounds of lookups (help, tasks, users)	Answer with task links or a ranked list
6	System	Saves the question and answer	Conversation can be reloaded later

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Example: At 09:10 on 03-Nov-2026 Asha (Employee) asks "What should I prioritise?". TFA lists 1. TSK-41 (Overdue), 2. TSK-52 (Today), 3. TSK-47 (Tomorrow, High), 4. TSK-49 (Tomorrow, Medium), 5. TSK-38 (Blocked, 4 days), each linking to its task page.

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Exceptions / alternate flows:

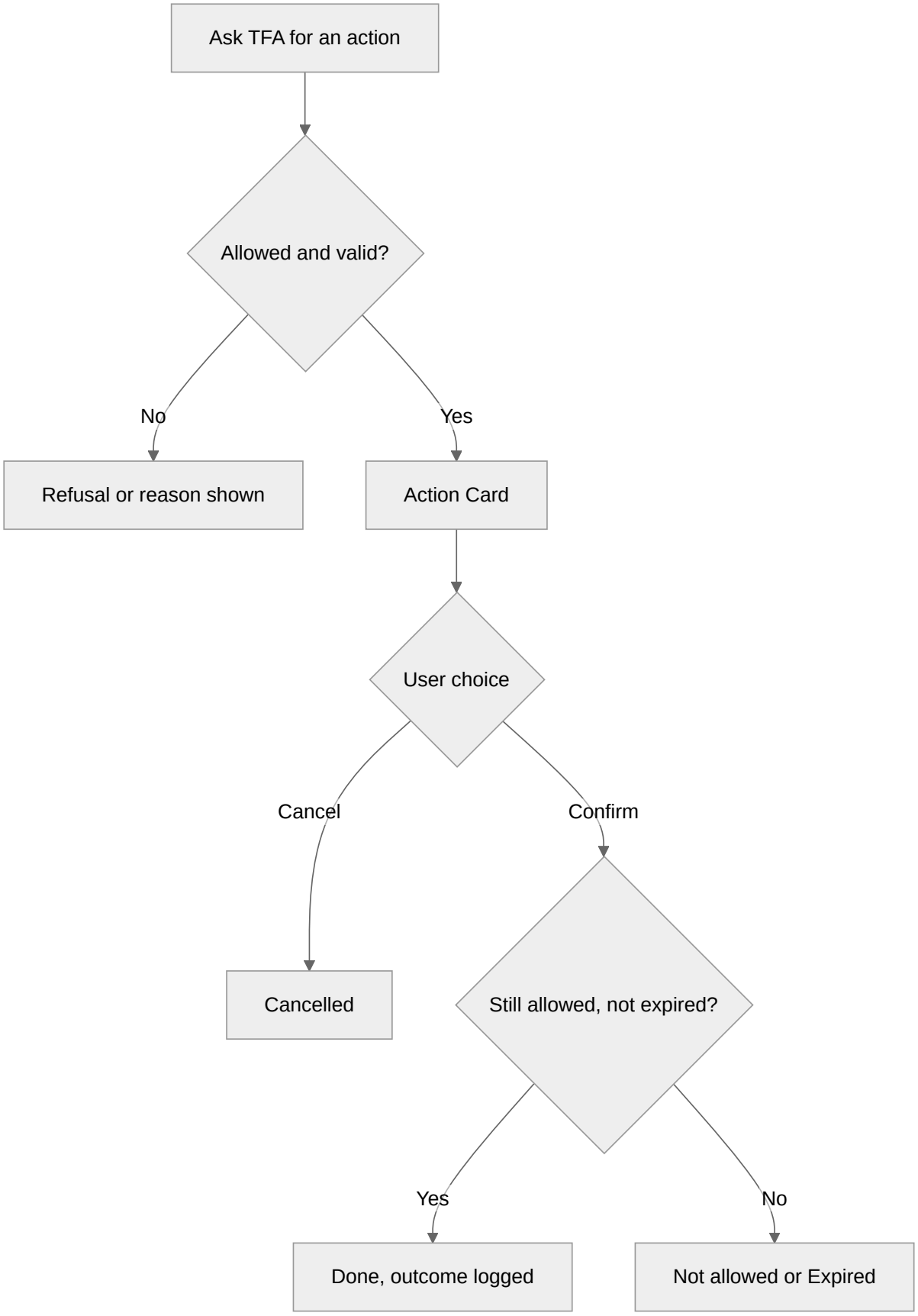
- No provider configured → "TFA is not available yet. An administrator needs to choose an AI model in TFA Settings."
- Provider fails (limit, bad key, timeout, unreachable, model missing) → plain message with **Retry** and **Use** another provider; nothing saved.

4.2 Confirm a Prepared Action

Trigger: User asks TFA to send a mail, change their name/email, or (administrator) change a role · **Actors:** User, System

Step	Actor / System	Action	Result
1	User	Asks, e.g. "Send a mail to my manager about TSK-12"	TFA resolves the recipient or asks who is meant
2	System	Checks permission and the module rules, shows an Action Card	"Nothing happens until you confirm." · expires in 10 minutes
3	User	Optionally edits (mail only), then clicks Confirm	Permission checked again
4	System	Runs the action once and records the outcome	"Done…" message, card shows **Done**

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Example: Ravi (Employee) asks "Send a mail to my manager about TSK-12". The task was created by Priya (Manager), so TFA shows a Mail draft to Priya with subject "Update on TSK-12". Ravi edits the message and clicks **Send mail**. TFA replies "Done. Your mail "Update on TSK-12" was sent to Priya Nair."

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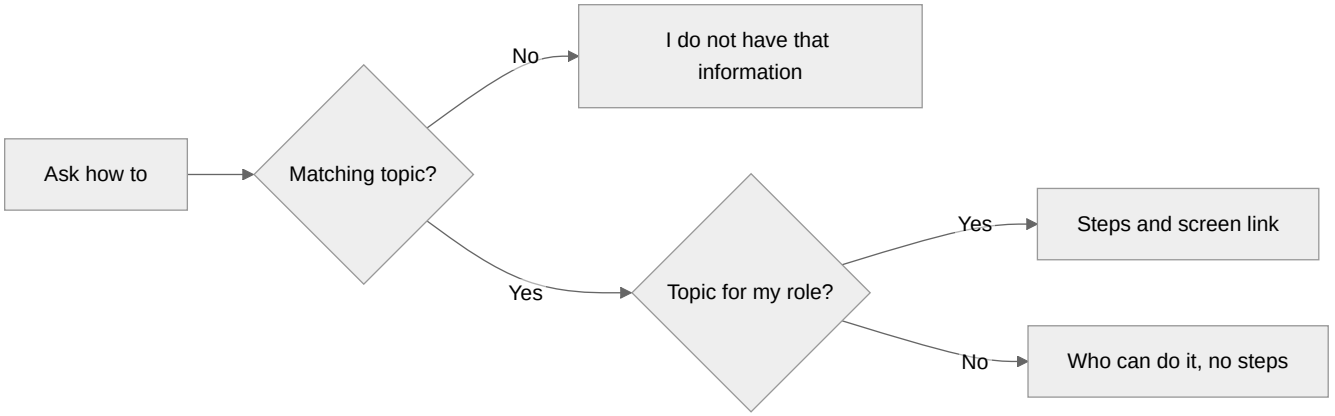
Exceptions / alternate flows:

- The card is older than 10 minutes → "This request has expired. Please ask TFA again."
- Confirm clicked twice → the action runs once; second click → "This action has already been done."
- Administrator asks to change their own role → "For safety I do not change your own role. Please ask another administrator to do it."

4.3 Get How-To Help

Trigger: User asks "How do I...?" · **Actors:** User, System

Step	Actor / System	Action	Result
1	User	Asks, e.g. "How do I change a task status?"	–
2	System	Searches the 19 help topics, uses up to 3 best matches	–
3	System	Topic for the user's role → steps and screen link; otherwise only who can do it	No invented features



Example: An Employee asks "How do I create a task?". TFA answers "Tasks are created by Managers and Administrators. If you need a new task, ask your manager to create and assign it to you."

Exceptions / alternate flows:

- No topic scores high enough → TFA says it does not have that information.

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5. Screen-by-Screen Requirements

SCREEN 1: TFA Panel (chat)

Field	Value
Screen name	TFA Panel
Module	TFA Assistant (08.1)
Access	Administrator, Manager, Employee (default); Custom role if granted `assistant.chat`

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Field	Value
Priority	Must

Purpose Let the user ask TFA questions and read its answers in a running conversation, from any signed-in screen.

Screen Elements

- Floating **TFA** button (bottom right) with voice hint.
- Header: "TFA – TaskFlow Assistant", **New chat** (+), **Close** (×).
- Model selector with **Local**/**Cloud** tag (menu opens only with 2 or more configured providers).
- Cloud notice with **I understand** and **Cancel**.
- Conversation feed: messages, "Permission required" bubble, notices, numbered ranked task list with tags (Overdue, Today, Tomorrow, `n` days, Blocked), screen link buttons, provider name under each answer.
- Welcome bubble and four starter suggestions on an empty chat.
- Message box "Ask TFA anything about TaskFlow...", microphone, **Send** (Enter sends, Shift+Enter new line).
- Footer: "TFA can make mistakes. Actions always need your confirmation."
- Provider failure card with **Retry** and **Use** another provider.

Functional Requirements

ID	Requirement	Priority
REQ-TFA-01	The system shall show the TFA button and panel on every signed-in screen only to users holding `assistant.chat`.	Must
REQ-TFA-02	The system shall reload the user's own latest conversation, up to the last 100 messages, when the panel opens.	Must
REQ-TFA-03	The system shall show another user's conversation or action to no one, reporting it as "not found".	Must
REQ-TFA-04	The system shall accept messages of 1 to 2000 characters.	Must
REQ-TFA-05	The system shall neither send nor store a message that looks like it contains a password or API key, and shall show a safety notice instead.	Must
REQ-TFA-06	The system shall answer task questions (upcoming, due today, overdue, all open) using the same access rules as Task Management.	Must
REQ-TFA-07	The system shall rank the user's open assigned tasks: overdue first (longest overdue first), then due today, then nearest due date, higher priority first on the same day; completed and cancelled tasks excluded, blocked tasks tagged Blocked.	Must
REQ-TFA-08	The system shall show the top 5 ranked tasks (up to 10 on request), each linking to its task page, with the order calculated by the system and not by the AI.	Must
REQ-TFA-09	The system shall list only providers configured by an administrator in the model selector, and keep the conversation when the model is switched.	Must
REQ-TFA-10	The system shall ask for acknowledgement before the first message to each cloud provider in a browser, and remember it in that browser.	Must
REQ-TFA-11	The system shall save nothing when a provider fails and offer **Retry** and another configured provider.	Must

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ID	Requirement	Priority
REQ-TFA-12	The system shall start a new conversation on **New chat**, keeping earlier messages stored but no longer using them as context.	Should
REQ-TFA-13	The system shall use at most the last 12 messages (about 6000 characters) of the current conversation as context, excluding notices.	Should
REQ-TFA-14	The system shall limit each answer to at most 4 rounds of lookups.	Should
REQ-TFA-15	The system shall treat task, comment and mail text that TFA reads as data, never as an instruction.	Must
REQ-TFA-16	The system shall clear the panel and load the new person's own conversation when a different user signs in on the same browser.	Must
REQ-TFA-17	The system shall use the application name and time zone from Application Settings in answers and in deciding what is due today.	Should

Validations & Messages

Situation	Rule / message
No provider (panel)	"TFA is not available yet. An administrator needs to choose an AI model in TFA Settings."
Chosen provider removed	"`{Provider}` is not configured. Choose another model, or ask an administrator to set it up."
Empty or too long	"A message must be between 1 and 2000 characters."
Password or key detected	"For your safety I did not send or keep that message: it looks like it contains a password or an API key. ..."
Provider rate limit	"`{Provider}`'s free limit or rate limit has been reached. Try again in about `{n}` seconds, or choose another model."
Provider rejected key	"`{Provider}` did not accept the API key. Ask an administrator to check the key in TFA Settings, or choose another model."
Provider timeout	"`{Provider}` took too long to answer. You can retry, or choose another model."
Ollama not running	"Ollama is not running or cannot be reached. Ask an administrator to start it, or choose another model."
Model missing	"The configured model is not available on Ollama. ..." / "The configured model is not available on `{Provider}`. ..."
No answer	"I could not come up with an answer. Please try again, or ask in a different way."
Other failure	"TFA could not answer. Please try again." · "Could not start a new chat."
Cloud notice	"`{Provider}` is a cloud provider. Your message and related task details are sent to it."
No permission	"You do not have the required permission to perform this action. ... Please contact your system administrator for assistance."

Expected Result Every permitted user can ask TFA about TaskFlow and their own work from any screen, with the model of their choice, and recover from provider problems with one click.

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SCREEN 2: How-to Help (inside the TFA Panel)

Field	Value
Screen name	TFA Panel – how-to answers
Module	TFA Assistant (08.1)
Access	Holders of `assistant.chat` (Administrator, Manager, Employee by default)
Priority	Must

Purpose Answer "how do I...?" questions only from TaskFlow's own written help.

Screen Elements

- TFA's written answer in the conversation.
- Link button to the matching screen (e.g. My Profile, Board, Mail).

Functional Requirements

ID	Requirement	Priority
REQ-TFA-18	The system shall answer how-to questions only from the 19 built-in help topics, using at most 3 best-matching topics.	Must
REQ-TFA-19	The system shall add a link to the screen named in the matching topic.	Should
REQ-TFA-20	The system shall give only "who can do it" (no steps) for topics not meant for the user's role: Administrator-only topics (5) and Manager-and-Administrator topics (4).	Must
REQ-TFA-21	The system shall say it does not have the information when no topic matches, and never invent screens or features.	Must

Validations & Messages

Situation	Rule / message
Restricted topic (default)	"`{Topic}` is not available to your role."
Restricted topic (example)	"Application settings are managed by an Administrator. Please contact your system administrator."
No matching topic	TFA says it does not have that information (wording by the AI model).

> ⚠️ To be confirmed: role-restricted topics are matched by role **name**, so a custom role always receives the "who can do it" note even when it holds the needed permissions.

Expected Result Users get consistent, accurate steps with a link, or a clear note on who can do the task.

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SCREEN 3: Action Card (inside the TFA Panel)

Field	Value
Screen name	Action Card
Module	TFA Assistant (08.1)
Access	Shown to holders of `assistant.chat`; buttons work for holders of `assistant.actions.confirm`; role change also needs `roles.manage` (Administrator by default)
Priority	Must

Purpose Show exactly what TFA prepared and let the user confirm, edit (mail only) or cancel it.

Screen Elements

- Title: **Mail draft**, **Update your profile** or **Change role**; "Expires in 10 minutes".
- Fields: To, Subject, message (mail); Name/Email old → new (profile); User and Role old → new (role).
- Buttons: **Send mail** / **Save changes** / **Change role**, **Edit** (mail only), **Cancel**; text "Nothing happens until you confirm."
- Mail edit form: To (email addresses), Subject, Message, **Save changes**, **Back**.
- Final status: Done, Cancelled, "Expired. Ask TFA again.", Not allowed, Not done.

Functional Requirements

ID	Requirement	Priority
REQ-TFA-22	The system shall let TFA only prepare an action; only the user's own Confirm click shall run it.	Must
REQ-TFA-23	The system shall check the user's permission when the action is prepared and again, with the current sign-in, when it is confirmed.	Must
REQ-TFA-24	The system shall expire a prepared action 10 minutes after it is created.	Must
REQ-TFA-25	The system shall run an action at most once, even when Confirm is clicked several times at the same moment.	Must
REQ-TFA-26	The system shall prepare internal mail drafts that follow the Mail rules and are sent from the user's own mailbox.	Must
REQ-TFA-27	The system shall resolve "my manager" as the task's creator when that is someone else, otherwise the single active Manager of the user's departments, and ask the user when none or several match.	Should
REQ-TFA-28	The system shall allow editing of mail drafts only, with recipients limited to active TaskFlow users and unknown addresses named.	Must
REQ-TFA-29	The system shall prepare a change to the user's own name and/or email only, following the My Profile rules.	Should
REQ-TFA-30	The system shall prepare a role change only for holders of `roles.manage`, to an existing role named exactly, and never for the requester's own role.	Should
REQ-TFA-31	The system shall tell a user whose role was changed that they were signed out and will use the new role at next sign-in.	Should

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ID	Requirement	Priority
REQ-TFA-32	The system shall find active users by name or email to address a mail.	Should
REQ-TFA-33	The system shall answer password questions with fixed guidance pointing to My Profile and never ask for or accept a password.	Must
REQ-TFA-34	The system shall tell the user when a request needs an administrator.	Could
REQ-TFA-35	The system shall mark a failed action **Not done** and explain why, with nothing sent or changed.	Must

Validations & Messages

Situation	Rule / message
Prepared	"I have prepared this for you. Please review it. Nothing happens until you confirm."
Mail sent	"Done. Your mail "`{subject}`" was sent to `{names}`."
Profile updated	"Done. Your profile was updated."
Role changed	"Done. `{name}` is now a `{role}`. They were signed out and will use the new role the next time they sign in."
Failed	"I could not do that: `{reason}`. Nothing was sent." / "Nothing was changed."
Expired	"This request has expired. Please ask TFA again."
Already done	"This action has already been done."
No longer pending	"This action can no longer be confirmed (`{status}`). Please ask TFA again." / "This action can no longer be changed. Please ask TFA again."
Edit non-mail	"Only a mail can be edited."
Unknown recipient	"These are not active TaskFlow users: `{addresses}`."
Mail rules	"Write a subject or a message before sending." / "Add between 1 and 50 recipients." / "'`{value}`' is not an email address." (subject max 200, message max 20,000 characters)
Own role	"For safety I do not change your own role. Please ask another administrator to do it."
Needs administrator	"This needs an administrator. Please contact your system administrator for assistance."
No permission (role)	"You do not have the required permission to perform this action. Changing a user's role can only be performed by an authorized administrator. ..."
Network failure	"Could not confirm that action." / "Could not cancel that action." / "Could not save your changes."

> ⚠️ To be confirmed: the fixed password guidance and the "Change your password" help topic still describe entering the **current password** and say only *other* sessions end. The built Change Password screen uses an authenticator code and signs the user out everywhere (My Profile BR-PROF-06, BR-PROF-08). The text should be aligned.

Expected Result Routine mails and own-profile changes are carried out with one confirmation, administrators change roles from the chat, and nothing happens without confirmation or beyond the

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user's permissions.

SCREEN 4: Voice Controls (inside the TFA Panel)

Field	Value
Screen name	TFA Panel – voice controls
Module	TFA Assistant (08.1)
Access	Holders of `assistant.chat` (Administrator, Manager, Employee by default), in Chrome or Edge
Priority	Could

Purpose Let users talk to TFA instead of typing and hear the answer.

Screen Elements

- Microphone button ("Speak a command (Chrome or Edge)").
- Listening bar: Waiting for "Hey TF assistant", Listening... say your command, Hands-free is off, Voice is paused; **Turn on "Hey TF assistant"** / **Turn on hands-free** / **Turn off**.
- "Heard: ..." text and voice hint under the TFA button.

Functional Requirements

ID	Requirement	Priority
REQ-TFA-36	The system shall open the panel and greet the user by first name ("Hi `{first name}`, how can I help you?") when the wake phrase "Hey TF assistant" is heard.	Could
REQ-TFA-37	The system shall send a spoken question like a typed message after about 2 seconds of silence, applying all chat rules.	Could
REQ-TFA-38	The system shall let the microphone button dictate a single command into the message box.	Could
REQ-TFA-39	The system shall read aloud answers to spoken questions only, up to 500 characters, pausing the microphone while speaking.	Could
REQ-TFA-40	The system shall remember a deliberate "Turn off" in the browser, and never turn voice off permanently because of an error.	Could
REQ-TFA-41	The system shall use the browser's own speech service and keep no recording.	Must

Validations & Messages

Situation	Rule / message
Browser without voice	"Voice input is not supported in this browser. Try Chrome or Edge, or type your question."
Microphone blocked	"Microphone access was denied. Allow it in your browser to use voice, or type your question."
No microphone	"No microphone was found. You can type your question instead."
No internet for voice	"Voice needs an internet connection in this browser. You can type your question instead."
Auto start blocked	"Click anywhere on the page to turn "Hey TF assistant" back on."

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Expected Result Users in Chrome or Edge can use TFA hands-free; typing always works as a fallback.

SCREEN 5: Background Safeguards (no screen)

Field	Value
Screen name	None – background safeguards
Module	TFA Assistant (08.1)
Access	Applies automatically to every TFA user (Administrator, Manager, Employee, Custom role); no screen
Priority	Must

Purpose Record what happened to every TFA action and stop any one user from using up the AI provider's quota.

Screen Elements

- None. Users only see the "too quickly" message.

Functional Requirements

ID	Requirement	Priority
REQ-TFA-42	The system shall allow at most 20 messages per user in any one-minute window, counting only messages that pass the length and safety checks.	Must
REQ-TFA-43	The system shall record each action outcome (prepared, confirmed, cancelled, expired, refused, failed) with user, action kind and time.	Must
REQ-TFA-44	The system shall never store message bodies, passwords or API keys in the audit record.	Must
REQ-TFA-45	The system shall record a refusal both when TFA refuses to prepare an action and when a confirm is refused.	Should

Validations & Messages

Situation	Rule / message
Over the limit	"You are sending messages too quickly. Please wait about `{n}` seconds and try again."

Expected Result Every action outcome is traceable without sensitive content, and no single user can flood TFA.

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6. Business Rules

Global rules (authentication, permission-based access, Own/Team/Organization scope, confirm-before-act) are in section 6 of the overall BRD (`../../00-Overall/BRD-TaskFlow-Overall.md`) and apply here.

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> ⚠️ To be confirmed: the exact global `BR-nn` numbers to cite here, once the overall BRD numbering is final.

ID	Rule	Applies to
BR-TFA-01	TFA can do only what the signed-in user could already do in TaskFlow; every tool uses the same permission rules as the screens.	SCREEN 1–3
BR-TFA-02	Every action needs the user's explicit confirmation; the AI model can never run an action.	SCREEN 3
BR-TFA-03	Passwords and API keys never pass through the chat: they are not sent to a provider and not stored.	SCREEN 1, 3
BR-TFA-04	A user works only with their own conversation and their own prepared actions.	SCREEN 1, 3
BR-TFA-05	Task answers never show a task the user could not open on the Task Management screens.	SCREEN 1
BR-TFA-06	The work order is calculated by the system, not by the AI, so it is the same for every model.	SCREEN 1
BR-TFA-07	How-to answers come only from the built-in help topics; TFA never invents screens or features.	SCREEN 2
BR-TFA-08	Prepared actions expire after 10 minutes and run at most once.	SCREEN 3
BR-TFA-09	Mail drafts follow the Mail rules; profile changes follow the My Profile rules; role changes follow Employee Management rules.	SCREEN 3
BR-TFA-10	TFA never changes the requester's own role and never changes a password.	SCREEN 3
BR-TFA-11	Only providers configured by an administrator can be used; cloud providers need the user's acknowledgement first.	SCREEN 1
BR-TFA-12	Each user may send at most 20 messages per minute.	SCREEN 5
BR-TFA-13	Every action outcome is audited without message bodies, passwords or keys.	SCREEN 5
BR-TFA-14	TaskFlow keeps no voice recording.	SCREEN 4

Change from original BRD

Original requirement	Current requirement	Reason
Not in original BRD (no assistant screen)	TFA Assistant added as a new sub-module with chat, actions, help, safeguards and voice	Added to reduce training effort and speed up routine work; AI provider setup added as TFA Settings

7. Non-Functional Requirements

Global NFRs are in section 7 of the overall BRD. Sub-module specifics:

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ID	Category	Requirement	Measure / target
NFR-TFA-01	Security	Secret screen on every message before any provider call	0 messages containing a detected password or key sent or stored
NFR-TFA-02	Security	Permission re-checked at confirm time	A permission removed after preparing makes the confirm fail
NFR-TFA-03	Privacy	A local provider option where nothing leaves the machine	Ollama selectable when configured
NFR-TFA-04	Reliability	Provider failures lose nothing	Retry resends the same text; no partial messages stored
NFR-TFA-05	Reliability	Concurrent confirms are safe	Duplicate confirm never runs an action twice
NFR-TFA-06	Fair use	Per-user message limit	20 messages per user per minute
NFR-TFA-07	Compatibility	Voice in browsers with speech recognition	Chrome and Edge; typing works in every supported browser
NFR-TFA-08	Usability	Every provider problem has a plain message and a next step	Each failure in SCREEN 1 offers Retry or another model

8. Data Requirements

Entity / data set	Purpose	Key fields	Retention / rules
Conversation	A user's chat thread	Owner, created, updated	Own only; kept indefinitely
Message	One question, answer, tool result or notice	Conversation, role, text, provider used, time	Last 100 shown; secrets never stored
Pending action	A prepared action waiting for confirmation	Owner, kind (Send mail, Update own profile, Change user role), details, status, created, expires	Expires after 10 minutes; resolved once
Audit record	Action outcome	User, kind, outcome, time	No bodies, passwords or keys
Help topics (19)	How-to knowledge	Title, roles, keywords, screen link, steps	Shipped with the application
Browser settings	Chosen model, cloud acknowledgement, hands-free off	Per browser	Not stored on the server

9. Integrations & External Dependencies

System / service	Purpose	Required?	Notes
Ollama (local AI)	Answers questions on the local machine	One provider required	"Ollama is not running..." when unreachable

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System / service	Purpose	Required?	Notes
Google Gemini (cloud AI)	Answers questions	Optional	Cloud notice; rate limit and key messages
OpenAI GPT (cloud AI)	Answers questions	Optional	Cloud notice; rate limit and key messages
Browser speech service	Speech to text and text to speech	Optional	Chrome, Edge; needs internet in some browsers
TaskFlow Mail, My Profile, Employee Management, Task Management	Carry out actions and lookups	Required	Same rules as their own screens
TFA Settings	Provider, model and API key configuration	Required	See `../../06-Settings/06.3-TFA-Settings/BRD-TFA-Settings.md`

10. Reporting Requirements

Not applicable for this sub-module. The audit log has no report or screen in this release.

11. Recommended Technology Stack (optional)

Not applicable for this sub-module. The stack is recorded once in the overall BRD and the TDD.

12. Assumptions, Constraints & Dependencies

Type	Description
Assumption	An administrator configures at least one AI provider before users rely on TFA
Assumption	Users understand that AI answers can be wrong ("TFA can make mistakes")
Constraint	The message limit is held in memory; a server restart resets it
Constraint	Voice depends on the browser's speech service (Chrome, Edge)
Dependency	Mail, My Profile, Employee Management and Task Management rules
Dependency	Application Settings for application name and time zone

13. Risks

ID	Risk	Impact	Likelihood	Mitigation
R-TFA-01	AI gives a wrong or invented answer	Medium	Medium	Help only from built-in topics; system-calculated ranking; footer warning
R-TFA-02	Prompt injection through task, comment or mail text	High	Low	Text treated as data; every action needs confirmation and a permission check

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ID	Risk	Impact	Likelihood	Mitigation
R-TFA-03	Company data sent to a cloud provider	High	Medium	Cloud notice; local Ollama option; secret screen
R-TFA-04	Conversations kept indefinitely with no self-service delete	Medium	Medium	⚠️ To be confirmed: retention period or "delete my history"
R-TFA-05	Provider quota exhausted by one user	Medium	Low	20 messages per user per minute
R-TFA-06	Help text out of step with built screens (password guidance)	Low	High	⚠️ To be confirmed: align texts with My Profile

14. Acceptance Criteria

ID	Criterion	Verified by
AC-TFA-01	A user without `assistant.chat` sees no TFA button; a user with it opens the panel and sees only their own last conversation.	SCREEN 1, REQ-TFA-01, REQ-TFA-02, REQ-TFA-03, REQ-TFA-16
AC-TFA-02	A 2001-character message is refused; "my password is Hunter2!" is neither sent nor stored.	SCREEN 1, REQ-TFA-04, REQ-TFA-05
AC-TFA-03	"What are my upcoming tasks?" lists only tasks the user can open, using the application time zone for "today".	SCREEN 1, REQ-TFA-06, REQ-TFA-15, REQ-TFA-17
AC-TFA-04	"What should I prioritise?" returns the same ranked top 5 with any model, with task links and tags.	SCREEN 1, REQ-TFA-07, REQ-TFA-08
AC-TFA-05	Only configured providers appear; the first cloud message shows the notice; switching model keeps the conversation.	SCREEN 1, REQ-TFA-09, REQ-TFA-10
AC-TFA-06	A provider timeout shows Retry and another provider, and nothing is stored.	SCREEN 1, REQ-TFA-11
AC-TFA-07	New chat empties the panel; earlier messages are not used as context; context and lookup limits are respected.	SCREEN 1, REQ-TFA-12, REQ-TFA-13, REQ-TFA-14
AC-TFA-08	An Employee asking how to create a task gets "who can do it" only; an Administrator gets steps and a link; an unknown topic gets "no information".	SCREEN 2, REQ-TFA-18, REQ-TFA-19, REQ-TFA-20, REQ-TFA-21
AC-TFA-09	A mail to "my manager" is prepared to the task creator, can be edited, and is sent only after Send mail.	SCREEN 3, REQ-TFA-22, REQ-TFA-26, REQ-TFA-27, REQ-TFA-28, REQ-TFA-32
AC-TFA-10	A card confirmed after 10 minutes shows Expired; a double click runs once; a removed permission makes confirm fail.	SCREEN 3, REQ-TFA-23, REQ-TFA-24, REQ-TFA-25
AC-TFA-11	An own name/email change applies the My Profile rules after Save changes.	SCREEN 3, REQ-TFA-29

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ID	Criterion	Verified by
AC-TFA-12	An Administrator changes Ravi to Manager and is told Ravi was signed out; a Manager is refused; own role is refused.	SCREEN 3, REQ-TFA-30, REQ-TFA-31
AC-TFA-13	A password question returns fixed guidance with a My Profile link; an admin-only request gets the "needs an administrator" text; a failed action shows Not done.	SCREEN 3, REQ-TFA-33, REQ-TFA-34, REQ-TFA-35
AC-TFA-14	In Chrome, "Hey TF assistant" opens the panel with a greeting; a spoken question is sent after silence and the answer is read aloud.	SCREEN 4, REQ-TFA-36, REQ-TFA-37, REQ-TFA-38, REQ-TFA-39
AC-TFA-15	"Turn off" is remembered after reload; no audio recording is stored by TaskFlow.	SCREEN 4, REQ-TFA-40, REQ-TFA-41
AC-TFA-16	The 21st message within a minute gets the "too quickly" message.	SCREEN 5, REQ-TFA-42
AC-TFA-17	Each prepared, confirmed, cancelled, expired, refused or failed action leaves one audit record without message text.	SCREEN 5, REQ-TFA-43, REQ-TFA-44, REQ-TFA-45

15. Requirements Traceability Matrix

Objective	Business rule(s)	Screen(s)	Functional requirement(s)	Acceptance criterion
OBJ-TFA-01	BR-TFA-04, BR-TFA-11	SCREEN 1	REQ-TFA-01, REQ-TFA-02, REQ-TFA-03, REQ-TFA-16	AC-TFA-01
OBJ-TFA-01	BR-TFA-11	SCREEN 1	REQ-TFA-09, REQ-TFA-10, REQ-TFA-11, REQ-TFA-12, REQ-TFA-13, REQ-TFA-14	AC-TFA-05, AC-TFA-06, AC-TFA-07
OBJ-TFA-02	BR-TFA-05, BR-TFA-06	SCREEN 1	REQ-TFA-06, REQ-TFA-07, REQ-TFA-08, REQ-TFA-17	AC-TFA-03, AC-TFA-04
OBJ-TFA-03	BR-TFA-02, BR-TFA-08, BR-TFA-09	SCREEN 3	REQ-TFA-22, REQ-TFA-24, REQ-TFA-25, REQ-TFA-26, REQ-TFA-27, REQ-TFA-28, REQ-TFA-29, REQ-TFA-32	AC-TFA-09, AC-TFA-10, AC-TFA-11
OBJ-TFA-04	BR-TFA-01, BR-TFA-10	SCREEN 3	REQ-TFA-23, REQ-TFA-30, REQ-TFA-31, REQ-TFA-33, REQ-TFA-34, REQ-TFA-35	AC-TFA-10, AC-TFA-12, AC-TFA-13
OBJ-TFA-04	BR-TFA-03	SCREEN 1	REQ-TFA-04, REQ-TFA-05, REQ-TFA-15	AC-TFA-02, AC-TFA-03
OBJ-TFA-04	BR-TFA-12, BR-TFA-13	SCREEN 5	REQ-TFA-42, REQ-TFA-43, REQ-TFA-44, REQ-TFA-45	AC-TFA-16, AC-TFA-17
OBJ-TFA-05	BR-TFA-07	SCREEN 2	REQ-TFA-18, REQ-TFA-19, REQ-TFA-20, REQ-TFA-21	AC-TFA-08
OBJ-TFA-06	BR-TFA-14	SCREEN 4	REQ-TFA-36, REQ-TFA-37, REQ-TFA-38, REQ-TFA-39, REQ-TFA-40, REQ-TFA-41	AC-TFA-14, AC-TFA-15

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16. Project Deliverables

#	Deliverable	Format
1	TFA panel, actions, help, safeguards and voice source code	Repository
2	Conversation, message, pending-action and audit data structures	Database migrations
3	19 built-in help topics	Markdown files shipped with the application
4	BRD, FRD, HLD, LLD for TFA	Markdown
5	Automated tests for chat, task ranking, actions, help topics, provider failures and the secret screen	Test projects and results

17. Glossary

Term	Meaning
TFA	TaskFlow Assistant, the built-in AI assistant
AI provider	The service that writes TFA's answers: Ollama (local), Google Gemini or OpenAI GPT (cloud)
Action Card	A card in the chat showing a prepared action that runs only on Confirm
Pending action	A prepared action waiting for confirmation; expires after 10 minutes
Work order	The system-ranked list of the user's open tasks
Wake phrase	"Hey TF assistant", which opens TFA hands-free
Secret screen	The check that blocks messages that look like passwords or API keys

18. Project Summary

TFA brings an AI assistant into every TaskFlow screen without giving up control. Users ask how TaskFlow works and get answers only from the official help, see a system-calculated "what to do first" list, and have mails and own-profile changes prepared for one-click confirmation; administrators can also change roles from the chat. Every action is permission-checked twice, confirmed by the user, run once and audited, while passwords and keys never pass through the chat and the organisation chooses between local and cloud AI.

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